

## **Project**

### **Smart Contract Applications**

#### **Problem Description:**

For each team you will have a problem to solve using Smart Contracts.

You will be given a link containing the smart contract and how it works and your task is to run this smart contract and write a report about it.

You can see which project you are assigned to from this

<https://docs.google.com/spreadsheets/d/1G2AsfjtYcC5PggyvixFaKnDfgt-YAo4FUItX1aq76kE/edit?usp=sharing>

#### **Report Specifications:**

1. The report should contain an explanation for every part of the code. That includes what does this mean and why it is used.
2. A Flow chart showing the flow of your smart contract.
3. Your Understanding of the problem you are required to solve and why this smart contract solves it.
4. Test cases that showcase your project and that the smart contract you have compiled and deployed is working as expected.
5. Logs of your transactions passing to the smart contracts and explaining what happens in your blockchain (Hint: Like the list of transactions you can see in etherscan).
6. (Bonus) Create a UI to showcase the dApp idea and interact with the smart contract using ethers.js as backend (ui can be a simple html/css web page).
7. (Bonus) Use static smart contract analyzers like Slither <https://github.com/crytic/slither> to scan your code , identify vulnerabilities and show their fixes .
8. (Bonus) Deploy and Run your code on a test net like Sepolia instead of using local network like ganache